

CPADN: Cross-Product Attention based Doubly-residual Network for Sales Forecasting

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ABSTRACT

Deep learning-based sales forecasting effectively captures temporal patterns in historical sales data. However, a major limitation in existing research is the frequent neglect of complex inter-product dependencies, despite the acknowledged impact of promotions and marketing on sales fluctuations. To address this issue, we propose the **Cross-Product Attention based Doubly-residual Network (CPADN)**, a novel deep learning framework with two key innovations. First, a time-sensitive **Cross-Product Attention** mechanism dynamically models inter-product dependencies and sales trends, while reducing model complexity via adaptive attention weights. Second, a **Doubly-residual Network** structure deeply fuses product-specific historical patterns with joint representations, enhancing multi-level feature extraction. We validated CPADN through extensive experiments on three public datasets: M5 Forecasting, SIDFC, and Favorita Forecasting. Considering the importance of lead time in supply chain decisions, we included forecasting scenarios with varying time lags for comparison. Results show CPADN consistently outperforms baseline methods, reducing average error by 5.2% on M5, 14.6% on SIDFC, and 3% on Favorita. Notably, CPADN exhibits strong stability and adaptability across short-, medium-, and long-term forecasts, with robustness confirmed under diverse data distributions and time horizons. These results affirm the practical value and theoretical contribution of CPADN in addressing complex sales forecasting challenges.

1. Introduction

In recent years, deep neural networks have achieved remarkable progress in computer vision, natural language processing, and time-series analysis, substantially improving the capability of modeling complex sequential data (Thakkar & Chaudhari, 2021). Sales forecasting in real-world retail scenarios is a cornerstone for inventory control, replenishment planning, and operational decision-making: inaccurate forecasts may lead to stockouts or overstock, thereby degrading service levels and increasing operational costs. Since sales records are naturally organized as time series, transferring deep learning models from the time-series forecasting literature to sales forecasting is a natural starting point (Chen & Ou, 2011). However, retail-specific characteristics—such as strong volatility, product interactions, and decision-related constraints—often prevent general-purpose time-series models from being “directly applied” with stable and reliable performance. Therefore, it remains important to develop forecasting frameworks that are better aligned with the underlying mechanisms of retail sales.

In the time-series forecasting literature, recurrent neural networks and their variants (e.g., LSTM) have been widely used to capture long-range temporal dependencies (Hochreiter & Schmidhuber, 1997; Salinas, Flunkert, Gasthaus & Januschowski, 2020; Fan, Sun, Yao, Zhang, Yan & Sun, 2021; Bandara, Bergmeir & Hewamalage, 2020; Sagheer & Kotb, 2019). Later, deep architectures built upon residual decomposition ideas (e.g., the forecast/backcast mechanism in N-BEATS) improved trainability and stability via iterative backcasting and refinement (Oreshkin, Carpo, Chapados & Bengio, 2019). More recently, Transformer-based models have demonstrated strong expressiveness for long-sequence modeling through self-attention, with continual advances in both accuracy and efficiency (Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser & Polosukhin, 2017; Wu, Xu, Wang & Long, 2021; Zhou, Zhang, Peng, Zhang, Li, Xiong & Zhang, 2021). Meanwhile, multivariate time-series forecasting has gradually shifted from modeling temporal dependence within individual series to jointly modeling temporal and inter-series dependence. Some approaches treat variables as tokens to explicitly capture cross-variable interactions, whereas others incorporate structured priors or graph-based modeling to represent inter-variable relations and improve system-level forecasting performance (Liu, Hu, Zhang, Wu, Wang, Ma & Long, 2023; Wang, Xu, Yang, Wu, Li, Xie & Chen, 2024). Despite these advances, existing methods primarily focus on general temporal dependence or relatively static inter-series relations, and they often under-address dynamic product interactions and business-driven forecasting constraints in retail sales.

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Although the aforementioned deep time-series models excel at capturing long-sequence temporal patterns and have begun to consider inter-variable dependence, they still cannot directly and sufficiently match the key characteristics of retail sales forecasting. This mismatch is mainly reflected in three aspects. First, retail sales typically exhibit strong short-term non-stationarity, where promotions, holidays, channel activities, and competitive strategies can trigger abrupt demand surges or drops within a short period (Ali, Sayin, Van Woensel & Fransoo, 2009). Second, multi-product demand in retail is rarely independent: substitution, complementarity, and within-category competition can substantially shape short-term sales trajectories, making models that rely only on single-series temporal patterns inadequate to fully describe real demand formation (Karray & Sigue, 2016). Third, sales forecasting often serves replenishment and stocking decisions and must account for lead time. In practice, a non-negligible temporal gap frequently exists between the decision time and the demand realization time, which makes the task closer to a gap/delay forecasting setting: the model is required to use a historical window $[T - M + 1, \dots, T]$ to predict $[T + D + 1, \dots, T + D + H]$, where D denotes the forecasting delay (gap). This setting reduces the availability of the most recent observations and increases extrapolation difficulty, thereby imposing stricter requirements on robustness and training stability (Guo, Ren, Du & Weng, 2025).

More importantly, product interactions in retail are not static but exhibit pronounced time sensitivity. For example, when a product runs a large-scale promotion within a specific time window, it may suppress the sales of similar products through price competition or attention diversion, or it may boost complementary products via bundling and cross-selling. When the promotion ends, or when stock becomes tight or a product is discontinued, such effects may quickly weaken, vanish, or even reverse. In other words, cross-product dependence evolves dynamically with time windows and market events. As a result, in multi-product sales forecasting, modeling temporal dependence alone—or relying solely on globally static correlations or fixed graph structures—often fails to reflect real market interactions. This calls for a forecasting framework that can explicitly model time-varying cross-product dependencies and remain stable under delay-aware forecasting settings.

To address these challenges, we propose a Cross-Product Attention based Doubly-residual Network (CPADN) for sales forecasting. CPADN introduces time-sensitive cross-product attention into a residual network and designs a forecasting decomposition module that splits the overall forecast into two complementary components: one focuses on temporal dependence of the target product, while the other captures time-varying cross-product dependence and interactions. We further adopt a forecast/backcast residual-decomposition principle to improve trainability and generalization stability in deep architectures. In addition, to align with replenishment decision-making, CPADN explicitly incorporates a forecasting delay D during training and evaluation by inserting a gap between the input and forecasting windows, enabling the model to learn stable mappings under realistic gap-forecasting scenarios.

In experiments, we conduct systematic evaluations of CPADN on three datasets from real-world sales forecasting scenarios and compare it with a range of representative baselines. The results show that CPADN achieves consistent improvements in both forecasting accuracy and stability, validating the effectiveness of time-sensitive cross-product modeling and residual decomposition for retail sales forecasting.

The key contributions of this paper are summarized as follows:

1. We present the first study on utilizing the time-sensitive cross-product attention mechanism for sales forecasting.
2. We design a block that can decompose sales forecasting into two partial forecasts which consider information of the target product and cross-product dependencies, respectively. The blocks are connected in a residual way.
3. We evaluate CPADN under a delay that better matches replenishment decisions, and demonstrate stable superiority over multiple baselines on real-world datasets..

This paper is organized as follows: Section 3 formulates the sales forecasting task, while Section 4 elaborates on the CPADN framework. Section 5 describes the experimental design, results, and analysis. Section 2 reviews related work and highlights our novel contributions. Finally, Section 7 summarizes key findings and outlines future research directions.

2. Related Work

Sales forecasting serves as a fundamental decision-making pillar in modern supply chain management, with its core value lying in constructing accurate future demand prediction systems for massive store-product combinations (Punia & Shankar, 2022). Currently, deep learning-based sales forecasting methods have evolved into two predominant research paradigms:

- Temporal dynamic modeling: Utilizing deep temporal neural networks (e.g., LSTM, TCN) to analyze the intrinsic patterns of product sales evolution over time, capturing long-term dependencies and periodic characteristics in sales data;
- Cross-product correlation modeling: Employing deep learning approaches such as graph neural networks (GNN) or attention mechanisms to deconstruct potential nonlinear relationships among products, uncovering synergistic and substitution effects between product categories.

In subsequent sections of this paper, we will systematically examine cutting-edge techniques closely related to our research, with in-depth analysis of their technical principles and application limitations.

2.1. Deep Learning for Temporal dynamic modeling

In response to the growing competitive pressure in the retail sector, businesses are leveraging predictive analytics to boost efficiency, streamline inventory management, lower costs, and drive profitability (Das & Ghosh, 2017).

Both traditional time series models and time series models based on deep learning are the main methods in sales forecasting, and their purpose is to mine the change patterns of observations with time. Traditional time series models, such as STL model Cleveland, Cleveland et al. (1990) and ARIMA model (Box & Cox, 1964; Cao, Wang, Duan, Zhang, Zhu, Huang, Tong, Xu, Bai, Tong et al., 2020) are classic methods to solve sales forecasting problems. STL model decomposes the stationary time series into trend information, seasonal information, and residual information of the series; ARIMA model is mainly to change the non-stationary time series into a stationary series through the difference method. Besides, the traditional machine learning methods, such as SVM (Jakkula, 2006), Xgboost (Chen & Guestrin, 2016) et al are also employed at sales forecasting. For example, Shen et al. (Shen, Wu, Yuan & Pei, 2018) utilized a range of models—including ARIMA, SVM, XGBoost, and Gaussian Processes (GP)—to predict product sales on one of China's largest e-commerce platforms, JD.com. While these methods have been widely studied and deployed, recent research indicates that simple statistical models often fall short in capturing the complex, nonlinear patterns present in modern sales data, making them insufficient for achieving satisfactory forecasting performance in many real-world retail scenarios.

Tackling time series forecasting by deep learning model has become extremely attractive research, since it has the powerful fitting ability and outstanding performance as the scale of data increases (Torres, Hadjout, Sebaa, Martínez-Álvarez & Troncoso, 2021), especially for the LSTM-based and Transformer-based model. LSTM model (Hochreiter & Schmidhuber, 1997) consists of recurrent cells whose states are affected by both past states and current input with feedback connections, and according to the 'gate' structure to improve the remembering capacity (Yu, Si, Hu & Zhang, 2019). Salinas et al. (Salinas, Flunkert, Gasthaus, & Januschowski, 2020) proposed the DeepAR model, which combines the AR model (Box, Jenkins, Reinsel & Ljung, 2015) with the LSTM model to obtain the probability distribution of future products sales. The MQRNN (Wen, Torkkola, Narayanaswamy & Madeka, 2017) employs the LSTM layer to encode time series, and according to two linear layers to summarize the encoding output and known future information separately as the decoder, combining these two parts as the prediction. The DFRNN (Wang, Smola, Maddix, Gasthaus, Foster & Januschowski, 2019) obtains the global factors by LSTM layer and random effect by noise LSTM layers, then, regards the global factors and random effect as distribution parameters to train the model. Although these models employ the statistic method to extract more information from LSTM layers output to improve the performance, they still capture the temporal information for each series independently.

The transformer model (Vaswani et al., 2017) is designed based on the attention mechanism, and it is another kind of deep learning model designed for sequence data. It can be modeled without considering the distance of input or output sequence, to reduce the information loss caused by a long sequence. Therefore, lots of Transformer-based models have been proposed. A tourism demand forecasting approach based on Attention-LSTM, incorporating neural network embeddings (Neural Networks embeddings) and K-means clustering for feature processing, significantly improves model performance (Kaya, Yilmaz, Yaslan, Ögüdücü & Çingir, 2022). Moreover, Autoformer (Wu, Xu, Wang, & Long, 2021) takes the Fast Fourier Transforms based on the Wiener–Khinchin theorem for the Query and Key in the attention mechanism, to predict by the highest similarity time interval. Informer (Zhou et al., 2021) proposed the ProbSparse Self-attention to ignore part of attention weights with lower contributions.

Although temporal modeling approaches have demonstrated efficacy in extracting historical sales dynamics for individual products, their fundamental assumption of product independence exhibits substantial divergence from the empirical realities of retail ecosystems.

2.2. Deep Learning for Cross-product correlation modeling

In the retail sector, product-level data typically exhibits characteristics of large scale and dynamic variation. With increasingly fierce market competition, various product combinations evolve rapidly due to market dynamics and multiple external factors (e.g., seasonality, promotional activities) (Ma & Fildes, 2021). Faced with such high-dimensional and non-stationary data characteristics, traditional methods relying on manual feature engineering and rule-based adjustments have become inadequate for modern retail scenarios. Therefore, there is an urgent need to adopt data-driven modeling approaches that automatically learn intrinsic data patterns to enhance forecasting performance (Böse et al., 2017).

In practical sales forecasting tasks, temporal dependencies and inter-product correlations often coexist interdependently. By fully exploiting information from relevant time series, it becomes possible to train models with better generalization capability and predictive accuracy while controlling model complexity (Wang et al., 2019). In recent years, academia has made significant progress in leveraging implicit product relationships to improve retail sales forecasting (Ma & Fildes, 2021). The field has witnessed significant technological evolution, which can be categorized into three distinct phases of development.

Early-stage research (2016-2018) primarily employed two-stage forecasting frameworks. A representative example is the work by Spiliotis et al. (Spiliotis, Makridakis, Semoglou & Assimakopoulos, 2022), who first classified products and then systematically evaluated 11 statistical methods and 7 machine learning approaches using data from 3,300 products of a Greek retail company. Similarly, Mejia and Aguilar (Mejía & Aguilar, 2024) proposed a clustering analysis method based on product characteristics, followed by selecting optimal forecasting models for each category. While these methods improved prediction accuracy, their segmented processing nature resulted in computational inefficiency and difficulty in capturing cross-category product relationships.

Mid-term research (2019-2021) began exploring end-to-end neural network architectures. Moraes et al. (de Castro Moraes, Yuan & Chew, 2024) innovatively proposed stacked (S-CNN-LSTM) and parallel (P-CNN-LSTM) hybrid models that effectively captured temporal dependencies and external features in retail data, outperforming traditional methods in both computational efficiency and forecasting accuracy. Zhao et al. (Zhao, Xu, Zhu, Liang, Zhang & Jiang, 2025) employed CNNs to extract spatial features combined with Bi-GRU networks to model temporal dependencies, achieving better forecasting performance through feature fusion. However, these CNN-RNN hybrid architectures suffered from model redundancy issues: RNNs struggled with long-sequence information processing, while CNNs' limited receptive fields resulted in insufficient spatial feature extraction.

Recent research (2022-present) has witnessed emerging Transformer and graph neural network-based approaches in multivariate time series forecasting. While not specifically designed for sales forecasting, these technical solutions provide potential references for retail sales prediction. For instance, iTransformer (Liu et al., 2023) employs self-attention modules between variables, treating individual time series as tokens to capture multivariate correlations through self-attention mechanisms. FCSTGNN (Wang et al., 2024) adopts a fully connected graph construction approach with mobile pooling convolution to maintain computational efficiency. TSMixer (Ekambaram, Jati, Nguyen, Sinthong & Kalagnanam, 2023) utilizes MLP modules between variables to capture complex inter-channel relationships represented by multi-level features extracted through fully connected layers.

Although these methods have been validated in different domains like power load forecasting and traffic flow prediction, their technical approaches for handling multivariate time series dependencies offer valuable insights for improving retail sales forecasting models.

3. Task Formulation

The task aims to predict future sales by analyzing the historical sales data of N products, and the set of products is represented as: $P = \{p^1, p^2, \dots, p^N\}$.

The historical sales data of each product p^j ($j = 1, 2, \dots, N$) defines its characteristics and is represented as a time series $\{s_1^j, s_2^j, \dots, s_T^j\}$, where s_t^j represents the sales value record of product p^j at time t .

We aim to estimate future sales for all products at time $T + D$, formally represented as $\{s_{T+D}^1, s_{T+D}^2, \dots, s_{T+D}^N\}$. Here, D is referred to as the *delay* in the forecasting task, representing the time interval between the current moment and the target future time point.

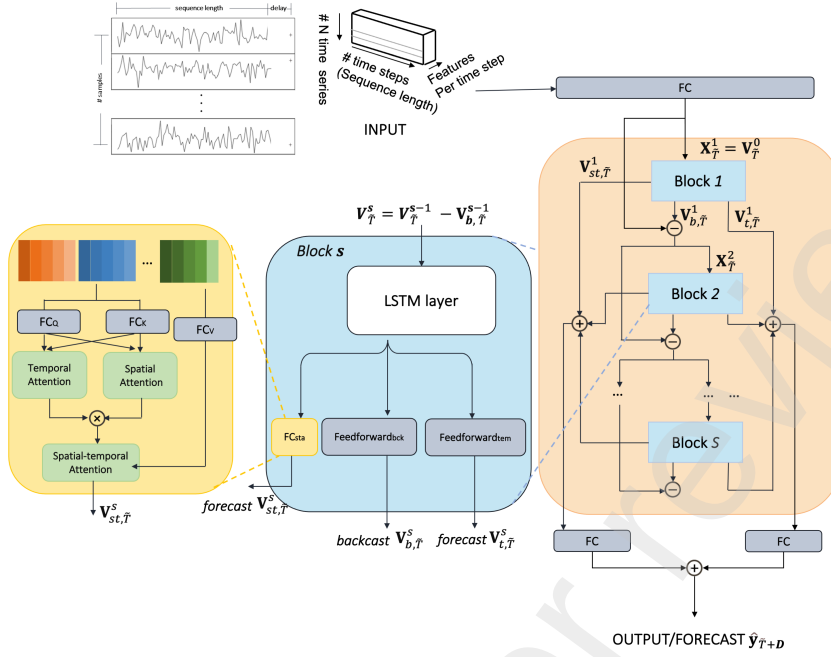


Figure 1: The network structure of *CPADN*. The part on the left side describes the whole network architecture in doubly residual stacking principle. The middle part depicts the structure in the basic building block. The part on the right side gives a introduction of cross-product forecast branch.

4. Our Methodology

Figure 1 presents the network architecture of *CPADN*, which comprises a series of homogeneous blocks interconnected via a doubly residual mechanism. Next, we discuss its design and functionality in detail of *CPADN*.

4.1. Model Input and Output

As shown in Figure 1, each sample can be represented as $(\mathbf{x}_{\tilde{T}}^j, y_{\tilde{T}}^j)$, where $\mathbf{x}_{\tilde{T}}^j$ is the feature vector of product p_j at time point $\tilde{T}$, reflecting its current sales status, and $y_{\tilde{T}}^j$ is the label associated with the sales forecasting target for p_j . Note that the sales forecasting task incorporates a *delay* of D , indicating that sales at time point $\tilde{T} + D$ are predicted using data available at the current time point $\tilde{T}$. In this setting, the actual sales of product p_j at time $\tilde{T} + D$ are represented by $s_{\tilde{T}+D}^j$, which serves as the label $y_{\tilde{T}}^j$.

In the *CPADN* model, $\mathbf{x}_{\tilde{T}}^j$ represents a fixed-length sequence comprising Historical sales records of p_j at time point $\tilde{T}$:

$$\mathbf{x}_{\tilde{T}}^j = [s_{\tilde{T}-M}^j, s_{\tilde{T}-M+1}^j, \dots, s_{\tilde{T}}^j], \quad (1)$$

where M is a hyperparameter controlling the length of the sequence in *CPADN*, determining the range of historical data to be analyzed. Since the *CPADN* aims to jointly analyze the sales data of N products to capture the dynamic temporal dependencies among them, the model input consists of the historical sales data of all products:

$$\mathbf{X}_{\tilde{T}} = [\mathbf{x}_{\tilde{T}}^1, \mathbf{x}_{\tilde{T}}^2, \dots, \mathbf{x}_{\tilde{T}}^N] \quad (2)$$

The primary goal of *CPADN* is to estimate the sales vector $s_{\tilde{T}+D}^j$ at the future time point $\tilde{T} + D$, expressed as:

$$s_{\tilde{T}+D} = \hat{\mathbf{y}}_{\tilde{T}} = F(\mathbf{X}_{\tilde{T}}, \Theta) \quad (3)$$

where $\hat{\mathbf{y}}_{\tilde{T}}$ represents the predicted sales results for all products at time point $\tilde{T} + D$, F denotes the forecasting process of the *CPADN* model, and Θ refers to the set of learnable model parameters.

4.2. Network Structure

In this subsection, we introduce the network architecture of *CPADN* and explain how it generates $\hat{\mathbf{y}}_{\tilde{T}}$ given $\mathbf{X}_{\tilde{T}}$ as input. *CPADN* employs homogeneous blocks, which are systematically arranged into stacks following the doubly residual stacking principle. This principle incorporates two residual branches: one refining the backcast predictions at each layer and the other enhancing the forecast branch. For time-series forecasting, the doubly residual stacking framework has demonstrated a clear advantage over alternative designs that omit backcast residual connections or restrict forward forecast links. In *CPADN*, we further extend the forecast branch to simultaneously integrate both single-product and cross-product forecasting.

As previously noted, each product's representation in $\mathbf{X}_{\tilde{T}}$ at a given time step consists solely of its sales figure at that specific moment. Consequently, $\mathbf{X}_{\tilde{T}}$ is structured as $\mathbb{R}^{N \times \tilde{T} \times 1}$. To enhance the feature representation, we first project it into a higher-dimensional space using a fully connected (FC) layer:

$$\mathbf{V}_{\tilde{T}}^0 = \mathbf{W}\mathbf{X}_{\tilde{T}} \quad (4)$$

where, $\mathbf{W} \in \mathbb{R}^H$ serves as a transformation matrix, mapping $\mathbf{X}_{\tilde{T}}$ to $\mathbf{V}_{\tilde{T}}^0 \in \mathbb{R}^{N \times \tilde{T} \times H}$.

Then, $\mathbf{V}_{\tilde{T}}^0$ undergoes transformation through L homogeneous blocks, producing the sequence $\mathbf{V}_{\tilde{T}}^0, \mathbf{V}_{\tilde{T}}^1, \dots, \mathbf{V}_{\tilde{T}}^L$. The l -th block accepts input $\mathbf{X}_{\tilde{T}}^l$ and outputs cross-product, single-product forecast and backcast representation:

$$(\mathbf{V}_{cp,\tilde{T}}^l, \mathbf{V}_{sp,\tilde{T}}^l, \mathbf{V}_{b,\tilde{T}}^l) = \text{Block}_l(\mathbf{X}_{\tilde{T}}^l). \quad (5)$$

Here $\mathbf{V}_{cp,\tilde{T}}^l, \mathbf{V}_{sp,\tilde{T}}^l, \mathbf{V}_{b,\tilde{T}}^l$ are all in space $\mathbb{R}^{N \times \tilde{T} \times H}$ and they will be used as *forecast* and *backcast* in the doubly residual topology.

As shown in Figure 1 (right), each block such as Block_l produces $\mathbf{V}_{cp,\tilde{T}}^l, \mathbf{V}_{sp,\tilde{T}}^l$ and $\mathbf{V}_{b,\tilde{T}}^l$. Here, we take advantage of doubly residual topology which is used commonly in building residual networks (Oreshkin, Carpo, Chapados, & Bengio, 2019; Tran, Mathews, Ong & Xie, 2021; He, Zhang, Ren & Sun, 2016), as it significantly enhances gradient flow and facilitates the efficient training of deep models. Specifically, $\mathbf{V}_{b,\tilde{T}}^l$ will be used for running a residual backcast connection by:

$$\mathbf{X}_{\tilde{T}}^{l+1} = \mathbf{X}_{\tilde{T}}^l - \mathbf{V}_{b,\tilde{T}}^l. \quad (6)$$

Otherwise, $\mathbf{V}_{cp,\tilde{T}}^l$ and $\mathbf{V}_{sp,\tilde{T}}^l$ act as partial forecast and are aggregated at the overall network level:

$$\hat{\mathbf{V}}_{sp,\tilde{T}}^A = \sum_{l=0}^L \mathbf{V}_{sp,\tilde{T}}^l \quad (7)$$

$$\hat{\mathbf{V}}_{cp,\tilde{T}}^A = \sum_{l=0}^L \mathbf{V}_{cp,\tilde{T}}^l \quad (8)$$

The final forecast will be relied on the aggregated forecast: $\hat{\mathbf{V}}_{cp,\tilde{T}}^A$ and $\hat{\mathbf{V}}_{s,\tilde{T}}^A$:

$$\hat{\mathbf{s}}_{\tilde{T}+D} = \mathbf{V}^{H \times 1} \hat{\mathbf{V}}_{sp,\tilde{T}}^{A'} + \mathbf{V}^{H \times 1} \hat{\mathbf{V}}_{cp,\tilde{T}}^{A'} \quad (9)$$

In the next subsection, we will introduce the detail architecture in each block.

4.3. Block Architecture

The architecture of each basic block (e.g., Block_{*l*}) is depicted in the middle part of Figure 1. In Block_{*l*}, an LSTM layer is followed by two classic feedforward networks and one feedforward network with cross-product attention mechanism. Let us denote the hidden output produced by the LSTM cell as:

$$\tilde{\mathbf{V}}_{\tilde{T}}^l = \text{LSTM}(\mathbf{X}_{\tilde{T}}^l). \quad (10)$$

Then, $\tilde{\mathbf{V}}_{\tilde{T}}^l$ is passed through three different branches. Two of them produce forecast $\mathbf{V}_{cp,\tilde{T}}^l$ and $\mathbf{V}_{sp,\tilde{T}}^l$ which represent the forecast considering the cross-product dependencies and cross-time relationships and one generate backcast $\mathbf{V}_{b,\tilde{T}}^l$. Specifically, two separate Feedforward neural networks are used to generate $\mathbf{V}_{sp,\tilde{T}}^l$ and $\mathbf{V}_{b,\tilde{T}}^l$, respectively:

$$\mathbf{V}_{sp,\tilde{T}}^l = \text{FeedForward}_{\text{tem}}^l(\tilde{\mathbf{V}}_{\tilde{T}}^l), \quad (11a)$$

$$\mathbf{V}_{b,\tilde{T}}^l = \text{FeedForward}_{\text{bkc}}^l(\tilde{\mathbf{V}}_{\tilde{T}}^l). \quad (11b)$$

Here, both $\text{FeedForward}_{\text{tem}}^l$ and $\text{FeedForward}_{\text{bkc}}^l$ are multi-layer perceptrons (MLPs) consisting of two dense layers:

$$\text{FeedForward}(\mathbf{X}) = \mathbf{W}_2 \text{RELU}(\mathbf{W}_1 \mathbf{X}) \quad (12)$$

Note that $\mathbf{V}_{sp,\tilde{T}}^l$ focuses on capturing temporal dependencies by using each product's own historical information for prediction and does not consider interactions between products. Therefore, CPADN proposes a cross-product attention mechanism to capture dynamic dependencies among different products.

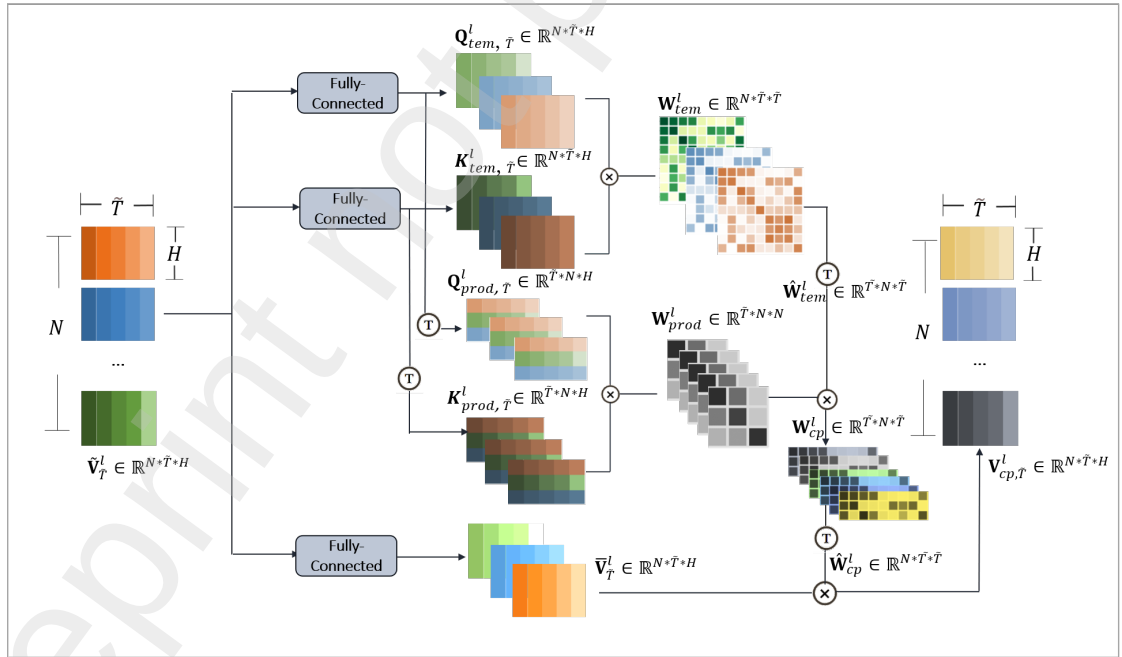


Figure 2: The illustration of cross-product attention mechanism used in CPADN.

As Figure 2 describes, the key idea of our cross-product attention mechanism originates from the extension of the self-attention mechanism (Bahdanau, Cho & Bengio, 2015). The input embeddings $\tilde{\mathbf{V}}_{\tilde{T}}^l$ are transformed into *Query*,

Key, and Value matrices using three sperate fully-connected layers. However, considering that the relationships between products may change over time, our cross-product attention mechanism uses *Query*, *Key* matrices to generate attention weights corresponding to the time dimension and the product dimension, respectively and finally merge them into dynamic cross-product attention. Specifically, CPADN first generates two key-query pairs $(\mathbf{Q}_{tem}, \mathbf{K}_{tem})$ and $(\mathbf{Q}_{prod}, \mathbf{K}_{prod})$. Then, time-sensitive cross-product attention weights are obtained by integrating temporal attention weights and product-level weights. $\tilde{\mathbf{V}}_{\tilde{T}}^l$ (i.e., the input of Block_{*l*}) is passed through two different full-connected networks to generate query $\mathbf{Q}_{tem, \tilde{T}}^l \in \mathbb{R}^{N \times \tilde{T} \times H}$, key $\mathbf{K}_{tem, \tilde{T}}^l \in \mathbb{R}^{N \times \tilde{T} \times H}$:

$$\mathbf{Q}_{tem, \tilde{T}}^l = \text{FC}_Q(\tilde{\mathbf{V}}_{\tilde{T}}^l) \quad (13a)$$

$$\mathbf{K}_{tem, \tilde{T}}^l = \text{FC}_K(\tilde{\mathbf{V}}_{\tilde{T}}^l) \quad (13b)$$

Then, the calculating temporal attention weights by adding a time masking with a lower triangular matrix $\mathbf{M}_{tem} \in \mathbb{R}^{\tilde{T} \times \tilde{T}}$ to avoid information disclosure in time dimension (Vaswani et al., 2017):

$$\mathbf{W}_{tem}^l = \text{Softmax}\left(\left(\frac{\mathbf{Q}_{tem, \tilde{T}}^l \mathbf{K}_{tem, \tilde{T}}^{lT}}{\sqrt{H}}\right) \cdot \mathbf{M}_{tem}\right) \quad (14)$$

Note that when process temporal attention $\mathbf{W}_{tem}^l$, the spatial dimension (i.e., cross-product dimension) is viewed as batch dimension to encourage $\mathbf{W}_{tem}^l$ describe relationships among different time points. Thus, for obtaining product-level weights, we could transpose the temporary and product dimension of $\mathbf{Q}_{tem, \tilde{T}}^l$ and $\mathbf{K}_{tem, \tilde{T}}^l$ to obtain the $\mathbf{Q}_{prod, \tilde{T}}^l \in \mathbb{R}^{\tilde{T} \times N \times H}$, $\mathbf{K}_{prod, \tilde{T}}^l \in \mathbb{R}^{\tilde{T} \times N \times H}$:

$$\mathbf{Q}_{prod, \tilde{T}}^l = \text{Transpose}(\mathbf{Q}_{tem, \tilde{T}}^l) \quad (15a)$$

$$\mathbf{K}_{prod, \tilde{T}}^l = \text{Transpose}(\mathbf{K}_{tem, \tilde{T}}^l). \quad (15b)$$

Then, the product-level attention to describing relationships among different time points can be obtained by

$$\mathbf{W}_{prod}^l = \text{Softmax}\left(\left(\frac{\mathbf{Q}_{prod, \tilde{T}}^l \mathbf{K}_{prod, \tilde{T}}^{lT}}{\sqrt{H}}\right) \cdot \mathbf{M}_{prod}\right) \quad (16)$$

Here $\mathbf{M}_{prod}$ is a mask matrix containing measurements for relationships among different products. Specifically, suppose the current status of a product is described by historical sales in recent one year. For products i and j , we calculate the Pearson-correlation coefficient between these two products as:

$$r_{i,j} = \frac{\sum_{t=1}^n (s_t^i - \bar{s}^i) (s_t^j - \bar{s}^j)}{\sqrt{\sum_{t=1}^n (s_t^i - \bar{s}^i)^2} \sqrt{\sum_{t=1}^n (s_t^j - \bar{s}^j)^2}}, \quad (17)$$

where $s_1^i, \dots, s_n^i$ ($s_1^j, \dots, s_n^j$) denotes the historical sales series of product i (j) in recent one year and $\bar{s}^i = \frac{1}{n} \sum_{t=1}^n s_t^i$ ($\bar{s}^j = \frac{1}{n} \sum_{t=1}^n s_t^j$). Here $r_{i,j}$ can be considered as a kind of similarity between product i and j . By assuming that there is an edge between product i and j if they are k -nearest neighbors and $r_{i,j}$ is larger than a threshold, the adjacency matrix is used to build spatial masking:

$$\mathbf{M}_{prod}^{ij} = \begin{cases} 1, & \text{if } (v_i, v_j) \in E \\ 0, & \text{otherwise} \end{cases} \quad (18)$$

where $(v_i, v_j) \in E$ indicates that there is an edge between product i and j .

Once we have temporal attention weights $\mathbf{W}_{tem}^l$ and spatial attention weights $\mathbf{W}_{prod}^l$, we calculate the time-sensitive cross-product attention weights as:

$$\mathbf{W}_{cp}^l = \mathbf{W}_{prod}^l * \hat{\mathbf{W}}_{tem}^l, \quad (19)$$

where $\hat{\mathbf{W}}_{tem}^l \in \mathbb{R}^{\tilde{T} * N * \tilde{T}}$ is obtained by transposing $\mathbf{W}_{tem}^l$.

Now, the time-sensitive cross-product embedding $\mathbf{V}_{cp, \tilde{T}}^l$ outputted by Block _{l} is obtained as follow:

$$\mathbf{V}_{cp, \tilde{T}}^l = \hat{\mathbf{W}}_{cp}^l * \bar{\mathbf{V}}_{\tilde{T}}^l, \quad (20)$$

where $\bar{\mathbf{V}}_{\tilde{T}}^l = \text{FC}_v(\tilde{\mathbf{V}}_{\tilde{T}}^l)$ is the values in the attention mechanism and $\hat{\mathbf{W}}_{cp}^l$ is obtained by transposing the first and second dimension of $\mathbf{W}_{cp}^l$ to match the size of $\bar{\mathbf{V}}_{\tilde{T}}^l$.

5. Experiments

5.1. Data Collection and Preparation

The empirical evaluation of the *CPADN* model employs three well-established public datasets from the sales forecasting domain.

The first dataset, **M5 Forecasting**¹, serves as an authoritative benchmark in the field. It contains daily sales records spanning from January 29, 2011 to April 24, 2016 for 3,049 Walmart products across three categories: household goods, food items, and leisure products. The dataset covers 10 stores located in three U.S. states, displaying sales patterns for both fast-moving consumer goods and durable goods with varying sales frequencies.

The second dataset, **Store Item Demand Forecasting Challenge (SIDFC)**², which comprises five years of daily sales data from January 1, 2013 to December 31, 2017 for 50 products sold across 10 stores. This dataset maintains a simple structure with only four attributes: sales date (excluding store closures and holidays), store ID, item ID, and daily sales volume. And all products sales exhibits strong trend characteristics.

The third dataset is **Favorita Forecasting**³. It includes nearly five years of daily sales records from January 1, 2013 to August 5, 2017 for 4,100 products. This dataset exhibits two particularly notable characteristics. First, it demonstrates significant sales imbalance that is typical of supply chain systems, with many items showing pronounced sparsity (over 50% of days with zero sales). Second, it captures the impact of the 2016 Ecuador earthquake, which caused abnormal sales patterns that persisted for several weeks following the event.

For preprocessing, the daily sales data in the **M5 Forecasting** and **SIDFC** datasets are aggregated into monthly time intervals. For the **Favorita Forecasting** dataset, the daily sales data are aggregated into weekly intervals. Additionally, all three datasets are subsequently divided into training, validation, and testing sets in a 7:2:1 ratio.

5.2. Experimental Setup

Forecasting Tasks: The evaluation of *CPADN* is conducted by varying the delay parameter (D , as defined in Section 3) to assess its performance across short-term, mid-term, and long-term forecasting tasks. For the **M5 Forecasting** and **SIDFC** datasets, for the delay D is $\{1, 3, 6, 12\}$. For the **Favorita Forecasting** dataset, D is set to $\{1, 2, 3, 4, 12, 24, 48\}$, a configuration that accommodates the weekly time granularity of this dataset while aligning with the monthly delays used in the other datasets.

Hyperparameter Selection: In *CPADN*, the input sequence length is determined by the hyperparameter M which is tailored to each dataset individually. Specifically, M is set to 12 for **M5 Forecasting**, 6 for **SIDFC**, and 12 for **Favorita Forecasting**. This setup ensures that the model utilizes historical sales data from the past year as the initial feature representation for each input sample. Besides, L representing the number of blocks in *CPADN*, is set to 3 for all configurations. Furthermore, when defining $\mathbf{M}_{prod}^{ij}$ in equation 18, we set $k=20\%$ and the threshold for the Pearson-correlation coefficient as 0.6.

¹<https://www.kaggle.com/competitions/m5-forecasting-accuracy/data>

²<https://www.kaggle.com/c/demand-forecasting-kernels-only>

³<https://www.kaggle.com/competitions/favorita-grocery-sales-forecasting>

Table 1

Performance on the M5 Forecasting dataset for varying delay values (D).

Delay	Error Metrics	Methodologies								
		CPADN	LSTMs	DeepAR	Nbeats	Seq2Seq	Informer	TSMixer	iTransformer	FCSTGNN
D=1	MAPE	0.2628	0.2822	0.2820	0.3659	0.4558	0.3098	0.4793	0.2966	0.5851
	MAE	113.5227	118.5161	120.2063	195.6874	145.5637	141.5707	154.6121	129.9877	180.2855
	RMSE	385.7793	407.1125	422.9337	832.8472	401.3959	565.0464	435.6178	491.9540	738.7406
D=3	MAPE	0.2938	0.3232	0.3189	0.3697	0.5068	0.3365	0.4586	0.3229	0.6086
	MAE	130.3916	139.2643	140.4281	196.3628	162.5372	160.2791	157.6508	148.3641	180.6156
	RMSE	417.1186	488.3187	497.6309	838.2632	425.7290	668.4596	434.0757	585.8088	757.4683
D=6	MAPE	0.3138	0.3414	0.3420	0.3740	0.5146	0.3581	0.4633	0.3342	0.4752
	MAE	138.2172	144.7382	149.5130	197.1741	164.0218	172.7542	157.2968	154.3008	167.3028
	RMSE	447.0111	501.7219	559.4978	839.7638	429.6365	675.3428	419.2254	603.8617	713.4331
D=12	MAPE	0.3330	0.3819	0.3815	0.3792	0.5560	0.3828	0.4505	0.3678	0.7084
	MAE	153.0121	170.9775	171.6820	198.4418	176.2869	191.3905	158.5807	176.9962	184.9242
	RMSE	509.5368	648.9565	673.6755	852.5730	457.1477	792.1984	399.3890	732.9276	840.3750
D=Mean	MAPE	0.3009	0.3321	0.3311	0.3722	0.5083	0.3468	0.4629	0.3304	0.5943
	MAE	133.7859	143.374	145.4574	196.9165	162.1024	166.4986	157.0351	152.4122	178.2820
	RMSE	439.8615	511.5274	538.4345	840.8618	428.4773	675.2618	422.0770	603.6380	762.5043

Table 2

Performance on the SIDFC dataset for varying delay values (D).

Delay	Error Metrics	Methodologies								
		CPADN	LSTMs	DeepAR	Nbeats	Seq2Seq	Informer	TSMixer	iTransformer	FCSTGNN
D=1	MAPE	0.1402	0.2346	0.2416	0.1787	0.2541	0.2740	0.1982	0.1228	0.1500
	MAE	2234.9733	3577.3567	3758.7733	2858.5267	3873.9767	3759.9800	2888.0667	2119.2967	2314.1800
	RMSE	2767.8494	4616.1237	4460.7339	3509.1214	5126.2233	5064.6215	3818.9771	2894.3431	3017.7851
D=3	MAPE	0.1444	0.2198	0.2192	0.1800	0.3248	0.2641	0.2222	0.1316	0.1553
	MAE	2344.8367	3505.1850	3507.8233	2906.6267	4436.9600	3677.1317	3332.3683	2233.1000	2440.8917
	RMSE	2834.8995	4325.2022	4252.4875	3459.8075	5788.2760	4827.5168	4220.1350	2954.7643	3218.6099
D=6	MAPE	0.0853	0.1222	0.1288	0.0821	0.2593	0.2417	0.1797	0.1203	0.1175
	MAE	1422.8367	2036.5617	2158.7983	1412.335	3981.8733	3542.7417	3019.8817	2122.4867	1792.9533
	RMSE	1957.5650	2662.7382	2965.4581	1862.0348	5285.5126	4794.1225	4206.2591	2748.8767	2475.5252
D=12	MAPE	0.0862	0.2529	0.2162	0.1688	0.2583	0.2472	0.1651	0.1412	0.1466
	MAE	1428.2133	3568.2000	3336.8250	2749.4717	4181.0017	3646.9767	2325.3400	2420.4533	2041.1550
	RMSE	1921.0845	4771.5128	4272.9653	3456.5549	5495.3777	4830.1953	3071.9339	3085.6887	3000.6194
D=Mean	MAPE	0.1140	0.2074	0.2015	0.1524	0.2741	0.2568	0.1913	0.128975	0.14235
	MAE	1857.7150	3171.8259	3190.5550	2481.7400	4118.4529	3656.7075	2891.41418	2223.83418	2147.295
	RMSE	2370.3496	4093.8942	3987.9112	3071.8797	5423.8474	4879.1140	3829.32628	2920.9182	2928.1349

Optimization Strategy: We employ RMSProp (Tieleman, 2012) as the optimizer for network training, using the mean squared error(MSE) as the loss function.

Performance Assessment: In line with prior studies (Wu, Xu, Wang, & Long, 2021) on sales forecasting and time-series regression, our evaluation relies on forecast accuracy, quantified by Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE):

$$MAPE = \frac{1}{n} \sum_i^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad (21)$$

$$MAE = \frac{1}{n} \sum_i^n |y_i - \hat{y}_i|, \quad (22)$$

$$RMSE = \sqrt{\left(\sum_{i=1}^n (y_i - \hat{y}_i)^2 \right)}, \quad (23)$$

where i corresponds to a test sample, while y_i and $\hat{y}_i$ represent the actual and predicted sales values, respectively.

Table 3

Performance on the **Favorita Forecasting** dataset for varying delay values (D).

Delay	Error Metrics	Methodologies								
		CPADN	LSTMs	DeepAR	Nbeats	Seq2Seq	Informer	TSMixer	iTransformer	FCSTGNN
D=1	MAPE	0.3026	0.3044	0.3139	0.4777	0.3193	0.3265	0.3529	0.3076	0.3934
	MAE	446.3184	472.3781	495.4341	983.2716	469.4832	514.1567	536.3978	460.3295	646.2469
	RMSE	1466.0569	1475.0051	1465.1536	2591.5494	1456.3655	1502.0396	1483.9552	1434.5732	1625.9046
D=2	MAPE	0.3144	0.3173	0.3323	0.4762	0.3373	0.3305	0.3734	0.3201	0.3934
	MAE	461.1979	483.3784	539.6074	981.6959	483.6652	510.2921	566.2684	482.3280	646.0978
	RMSE	1457.6287	1475.2807	1501.9182	2587.6813	1465.1258	1443.5970	1532.8774	1443.9972	1625.6726
D=3	MAPE	0.3221	0.3247	0.3404	0.4759	0.3447	0.3390	0.3878	0.3315	0.3933
	MAE	479.1354	500.1048	554.8250	983.6193	503.1600	534.8800	572.8442	497.3751	646.0225
	RMSE	1404.8998	1452.2304	1520.2954	2590.6180	1454.5912	1474.7844	1548.1138	1449.5689	1625.5747
D=4	MAPE	0.3257	0.3346	0.3468	0.4777	0.3518	0.3472	0.3907	0.3340	0.5453
	MAE	485.2045	535.0276	570.7152	985.2714	523.5379	547.9073	579.8834	491.7429	757.1530
	RMSE	1463.5470	1528.6590	1557.6780	2594.2263	1489.1853	1472.8127	1543.6348	1424.7796	1804.6310
D=12	MAPE	0.3362	0.3427	0.3528	0.4790	0.3784	0.3515	0.3993	0.3412	0.3916
	MAE	507.4942	558.8478	566.5653	976.8175	548.5548	543.8557	594.9285	527.9631	642.0422
	RMSE	1430.6078	1541.2665	1555.8529	2575.4887	1520.2146	1528.1097	1537.8241	1512.7606	1617.2470
D=24	MAPE	0.3962	0.3935	0.4013	0.4844	0.4126	0.3920	0.4906	0.3845	0.3900
	MAE	560.2238	661.5981	652.4268	985.3712	620.1565	618.5107	668.3776	634.0357	639.2132
	RMSE	1490.9199	1722.5241	1685.8836	2579.3529	1630.7359	1590.6264	1670.7475	1660.4190	1611.2508
D=48	MAPE	0.4094	0.3888	0.3902	0.4724	0.3810	0.3919	0.3983	0.3816	0.3846
	MAE	616.7942	624.9778	641.8434	1028.8366	642.3298	654.8873	593.6304	629.8074	627.6418
	RMSE	1573.7320	1620.2173	1665.6395	2713.3635	1583.0690	1635.8420	1532.3083	1606.2128	1590.6956
D=Mean	MAPE	0.3162	0.3203	0.3334	0.4769	0.36370	0.3358	0.3990	0.3429	0.41309
	MAE	467.9641	497.7222	540.1454	983.4646	555.3339	526.8090	587.4757	531.9402	657.7739
	RMSE	1448.0331	1482.7938	1511.2613	2591.0188	1498.2612	1473.3084	1549.9230	1504.6159	1642.9967

Table 4

Performances with/without Spatiotemporal modules at CPADN for varying delay D on **M5 Forecasting**, **SIDFC**, **Favorita Forecasting** dataset.

Delay	Error Metrics	M5 Forecasting			SIDFC			Delay	Favorita Forecasting		
		CPADN	B1	RN	CPADN	CPADN B1	RN		CPADN	B1	RN
D=1	MAPE	0.2775	0.2741	0.2628	0.1603	0.1490	0.1402	D=1	0.3134	0.3107	0.3026
	MAE	115.1761	115.5869	113.5227	2471.1283	2342.2850	2234.9733		490.7399	466.3841	446.3184
	RMSE	379.0218	401.7827	385.7793	3158.4219	2912.4550	2767.8494		1464.9453	1467.3033	1466.0569
D=2	MAPE	0.2949	0.2941	0.2809	0.1856	0.1610	0.1547	D=2	0.3243	0.3178	0.3144
	MAE	127.8045	126.0782	124.0511	2910.6633	2577.3983	2498.6483		507.7209	467.6551	461.1979
	RMSE	432.3915	429.8610	406.6614	3502.6220	3081.7935	2976.5487		1462.4708	1439.1981	1457.6287
D=3	MAPE	0.3050	0.3008	0.2938	0.1728	0.1520	0.1444	D=4	0.3327	0.3297	0.3257
	MAE	134.7401	132.9741	130.3916	2731.7267	2456.0833	2344.8367		537.5586	503.7123	485.2045
	RMSE	459.0688	429.6123	417.1186	3447.4128	2989.1526	2834.8995		1474.2939	1443.7128	1463.547
D=4	MAPE	0.3091	0.3084	0.3057	0.1663	0.1507	0.1602	D=8	0.3326	0.3386	0.3304
	MAE	138.8001	136.9327	135.5821	2652.3183	2396.4683	2578.7117		535.5580	516.1913	504.4127
	RMSE	481.4668	442.6584	427.521	3233.4033	2897.7196	3032.911		1517.3205	1502.5958	1449.0636
D=5	MAPE	0.3192	0.3195	0.3158	0.1383	0.130	0.1267	D=12	0.3398	0.3362	0.3362
	MAE	141.0612	140.4051	137.234	2204.0483	2087.4033	2037.0883		538.0168	516.7622	507.4942
	RMSE	488.2466	465.1825	429.8289	2802.6740	2765.1121	2687.0543		1457.4307	1441.9026	1430.6078
D=6	MAPE	0.3195	0.3156	0.3138	0.1146	0.1115	0.0853	D=24	0.3880	0.3949	0.3962
	MAE	142.5584	139.1557	138.2172	1878.1483	1842.3367	1422.8367		650.9778	570.2225	560.2238
	RMSE	494.8097	455.3095	447.0111	2496.1120	2451.949	1957.565		1895.8711	1510.92	1490.9199
D=Mean	MAPE	0.3042	0.3021	0.2955	0.1563	0.1424	0.1353	D=Mean	0.3385	0.3380	0.3343
	MAE	133.3567	131.8555	129.8331	2474.6722	2283.6625	2186.1825		543.4287	506.8213	494.1419
	RMSE	455.8342	437.4011	418.9867	3106.7743	2849.6970	2709.4713		1545.3887	1467.6054	1459.6373

5.3. Competitors

The evaluated approaches include

CPADN: Our approach;

LSTMs: LSTM (Long Short-Term Memory) networks are deep neural network architectures composed of multiple stacked LSTM units. As one of the most classical neural network models for processing sequential data, the number of network layers and hyperparameter settings must remain consistent to ensure the validity and comparability of model performance comparisons;

DeepAR (Salinas, Flunkert, Gasthaus, & Januschowski, 2020): This advanced time-series forecasting model is extensively utilized in real-world platforms such as Amazon SageMaker. It specializes in probabilistic forecasting and effectively processes numerous correlated time series simultaneously.

N-BEATS (Oreshkin, Carpo, Chapados, & Bengio, 2019): Designed with a unique bidirectional residual mechanism, this model integrates a backward link for constructing the residual architecture and a forward link that applies multiple linear neural networks to extract time-domain features

Seq2Seq (Vallés-Pérez, Soria-Olivas, Martínez-Sober, Serrano-López, Gómez-Sanchís & Mateo, 2022): A transformer-based model recently applied to sales forecasting. In Seq2Seq, the softmax function is removed from the attention mechanism, as sales forecasting is a regression task rather than a classification task, making the model more suitable for continuous predictions.

Informer (Zhou et al., 2021): This advanced Transformer-based architecture is designed for long-range time-series forecasting. To improve efficiency, it leverages the ProbSparse self-attention mechanism for complexity reduction and applies self-attention distillation to enhance temporal feature representation.

TSMixer (Ekamparam et al., 2023): This a lightweight time series forecasting model based on fully connected layers (MLPs), which efficiently captures cross-variable dependencies and temporal patterns in multivariate time series through multi-layer feature interaction and residual connections.

iTransformer (Liu et al., 2023): The model captures time series correlations by transposing the input dimensions of time series data based on the attention mechanism, while leveraging residual connections to progressively update temporal embeddings.

FCSTGNN (Wang et al., 2024): This model incorporates fully-connected graph convolution with a moving-pooling GNN layer to effectively capture spatiotemporal patterns and learn high-fidelity representations.

CPADN_B1: This model is derived from *CPADN* by isolating the residual framework. The remaining structure forms a new baseline model, which we refer to as **CPADN_B1**. This model uses a single time-sensitive cross-product attention mechanism.

RN: The time-sensitive cross-product attention mechanism is extracted from *CPADN* to form a new baseline. The remaining architecture is a residual network, which we denote as *Doubly-residual Network (DN)*.

5.4. Experimental Results

This section presents a comprehensive evaluation of *CPADN*'s performance in multi-step sales forecasting across different time horizons (delay intervals). We systematically assess the model's prediction accuracy, robustness, and temporal sensitivity through three key aspects.

5.4.1. Predictive Performance Across Delay Settings

We conducted a systematic evaluation of the *CPADN* model on three public datasets. As shown in Tables 1, 2, and 3, *CPADN* demonstrated stable performance advantages across all forecasting horizons (short/medium/long term).

In the **M5 Forecasting** dataset:

- *CPADN* reduced prediction error by 8.9% in MAPE compared to the suboptimal model iTransformer
- Achieved 6.7% lower MAE than LSTMs
- Notably ranked second in RMSE, only behind TSMixer

For the **SIDFC** dataset, *CPADN* achieve the best performance, the details as follow:

- 11.6% reduction in MAPE versus the suboptimal model iTransformer
- 13.5% improvement in MAE over the suboptimal model FCSTGNN
- Significant 18.8% decrease in RMSE compared to the suboptimal model iTransformer

For the **Favorita Forecasting** dataset, *CPADN* achieve the best performance, the details as follow:

- 11.6% reduction in MAPE versus the suboptimal model iTransformer
- 13.5% improvement in MAE over the suboptimal model FCSTGNN
- Significant 18.8% decrease in RMSE compared to the suboptimal model iTransformer

5.4.2. Model Robustness Verification

The Tables Table 4 outlines the ablation study results.

The experimental results demonstrate that the **CPADN** model exhibits significant advantages compared to the **RN** model. In the **M5 Forecasting** dataset, **CPADN** achieves error reductions of 5.3% in MAPE, 1.5% in MAE, and 4.2% in RMSE. For the **SIDFC** dataset, the error reductions reach 5.0% in MAPE, 4.3% in MAE, and 4.9% in RMSE. In the **Favorita Forecasting** dataset, **CPADN** shows error reductions of 1.1% in MAPE, 2.5% in MAE, and 0.5% in RMSE. Across all three datasets, the average performance improvements are 3.8% reduction in MAPE, 2.8% in MAE, and 3.2% in RMSE. These results fully confirm that the **CPADN** model significantly outperforms the **RN** model, validating the effectiveness of its time-sensitive cross-product attention mechanism.

Further comparison between **CPADN** and its **CPADN_B1** variant reveals equally notable differences. In the **M5 Forecasting** dataset, **CPADN** demonstrates error reductions of 6.5% in MAPE, 2.6% in MAE, and 8.1% in RMSE. For the **SIDFC** dataset, the improvements are more substantial with 13.4% reduction in MAPE, 11.7% in MAE, and 12.8% in RMSE. In the **Favorita Forecasting** dataset, **CPADN** achieves error reductions of 1.2% in MAPE, 9.1% in MAE, and 5.5% in RMSE. The averaged results across all three datasets show that **CPADN** reduces errors by 7.0% in MAPE, 7.8% in MAE, and 8.8% in RMSE. This comprehensive comparison not only reaffirms **CPADN**'s superiority over the **CPADN_B1** model, but also highlights the crucial role of the dual-residual structure in enhancing model performance.

6. Discussion

This study proposed *CPADN* model has demonstrated robust predictive performance through extensive experimental evaluation. Built upon deep learning foundations, *CPADN* leverages the low-complexity temporal-product interaction representation enabled by the temporally-sensitive cross-product attention mechanism, while incorporating the doubly-residual structure's capability to integrate historical variation patterns with joint modeling representations and perform deep-level feature enhancement. These synergistic components collectively strengthen the model's feature extraction capacity.

The temporally-sensitive cross-product attention mechanism represents an innovative attention approach that demonstrates remarkable advantages in jointly modeling inter-product dependencies and historical sales trends. Through its meticulously designed cross-dimensional weighting system, this mechanism achieves efficient representation of temporal-product interaction information. Its innovation primarily manifests in three aspects: First, it preserves the strengths of conventional attention mechanisms in temporal data representation; second, it overcomes the technical limitation of traditional methods that require segmented processing of product dependencies and sales trends; third, it innovatively incorporates a dynamic mask matrix based on Pearson correlation coefficients, which adaptively adjusts inter-product dependency weights to significantly enhance the model's adaptability to diverse product data. This design not only maintains computational efficiency but also substantially improves both accuracy and stability in sales forecasting tasks.

In the field of time series data analysis, the doubly residual structure has been widely adopted due to its exceptional processing capabilities, primarily stemming from its ability to effectively decompose time series data into trend and cyclical components. In current deep learning research, the application of doubly residual structures is typically confined to scenarios requiring simultaneous capture of both components. Notably, traditional deep learning-based product sales forecasting methods often study product historical sales patterns and joint modeling representations separately, fundamentally because conventional joint modeling representations require overly complex network architectures. Through the innovative proposal of a low-complexity joint modeling representation method, this study successfully achieves effective integration of the doubly residual structure with historical variation features. This integration not only profoundly enhances representational capacity but, more importantly, significantly improves the model's ability to represent and adapt to diverse sales data patterns.

The experimental results presented in Tables Tables 1, 2, and 3 demonstrate that while comparative models such as **TSMixer**, **iTransformer**, and **FCSTGNN** occasionally outperform *CPADN* in specific scenarios, *CPADN* maintains statistically superior overall performance across all three benchmark datasets. Particularly noteworthy is the fact that these test datasets exhibit distinct characteristics: the **M5 Forecasting** dataset combines both high-frequency and low-frequency product data, the **SIDFC** dataset displays pronounced periodicity features, and the **Favorita Forecasting** dataset exhibits significant imbalanced distribution with numerous outliers. *CPADN* consistently delivers robust performance across these diverse datasets, substantiating both the theoretical significance and practical applicability of our proposed methodology. Although minor performance gaps exist in certain individual metrics,

CPADN's comprehensive advantages in real-world business contexts establish it as the most reliable solution currently available, fully meeting practical application requirements.

This study aspires to invigorate the development of related fields through theoretical innovations and methodological breakthroughs, ultimately translating into tangible productivity and creating sustainable social value. It is our sincere hope that these research outcomes will not only advance academic progress but also make substantive contributions to industrial development and social well-being.

7. CONCLUSION AND FUTURE WORK

Our research presents *CPADN*, an innovative deep learning framework for sales forecasting that combines a temporally-sensitive cross-product attention mechanism with a dual residual structure. This novel architecture jointly models inter-product dependencies and historical sales trends while maintaining computational efficiency through its low-complexity design. The cross-product attention mechanism effectively captures complex product relationships while overcoming traditional complexity challenges, and the dual residual structure enables deep feature enhancement through integrated historical pattern analysis. Extensive validation across three diverse datasets demonstrates *CPADN*'s superior performance in short, medium and long-term forecasting scenarios compared to existing state-of-the-art approaches.

From a managerial perspective, our work highlights the value of treating sales data as a complex system, where joint modeling of product relationships and temporal patterns yields more accurate market insights. The framework's ability to extract multi-dimensional features provides businesses with enhanced predictive capabilities for data-driven decision making.

Looking ahead, we identify several promising research directions to further advance *CPADN*'s capabilities. First, we will enhance model interpretability through visualization techniques and attention weight analysis, improving transparency for business users. Second, we plan to incorporate external factors like pricing dynamics, climate impacts, and promotional effects through adaptive multi-source data fusion. Finally, we will investigate advanced architectures including graph neural networks and self-supervised learning to better capture high-dimensional, nonlinear relationships. These developments aim to establish *CPADN* as a next-generation decision support tool that delivers both predictive accuracy and actionable business intelligence across diverse market conditions.

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